

The biochemistry of Body fluids



First stage

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Body Fluids

- Total amount of fluid in the human body is approximately 60% of body weight
- Body fluid has been divided into two compartments
 - **Intracellular fluid (ICF)**
 - Inside the cells
 - 66.7% of total body water
 - **Extracellular fluid**
 - Outside the cells
 - 33.3% of total body water

Table(1) regulation of water balance

Daily input	Daily output
Drinking 1000ml	Urine 1000-1500ml
Eating 700-1000 ml	Sweat 400 ml
Water produced by metabolism 300-500ml	Respiration 400ml
	Faeces 200ml
Total 2000-2500ml	Total 2000-2500 ml

Disturbance of water balance: *Dehydration (Hypovolemia):*

Loss of water from the body in excess amounts leads to dehydration.

First the plasma becomes concentrated followed by other ECF and later ICF.

When water comes out of the cells, it also passes into ECF in exchange of K & Na passes into ICF from ECF. Loss of more than 20% of body water results in death.

Causes of dehydration:

1-Severe diarrhea and vomiting



2-Nonavailability of drinking water as in the case of desert.

3-Difficulty in swallowing and state of unconsciousness.

4-Loss of fluid from skin in case of burns.

5-Diabetic inspidus (ADH deficiency-polyuria)

6-Heart stroke

Water excess (hydration) hypervolemia:

The condition in which body water content is increased.



Causes:

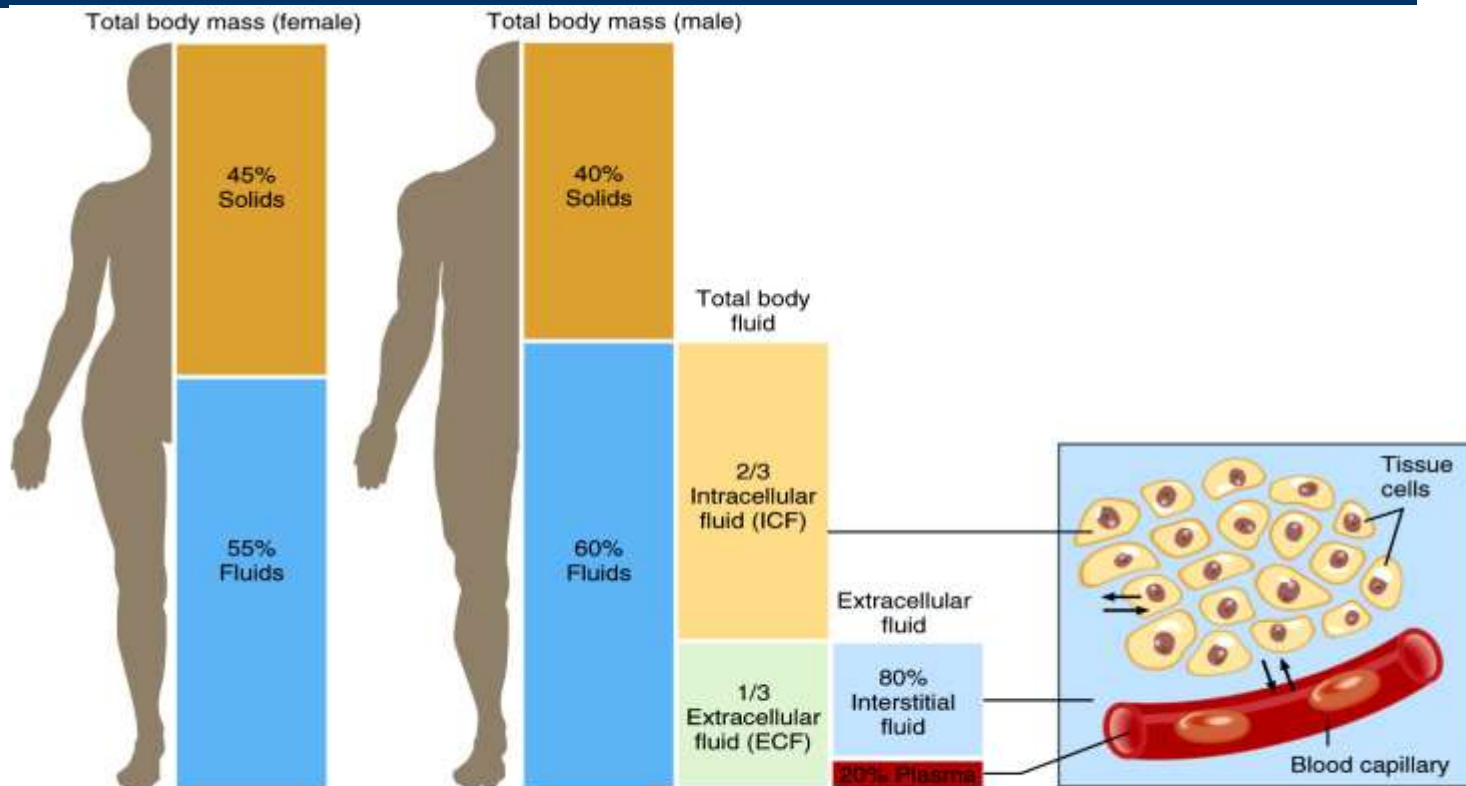
- 1-Hypersecretion of ADH
- 2- Following the administration of anesthetics his effect takes for about 12-36 hour after the surgery.
- 3-Renal failure.

Body Fluid Compartments

Extracellular fluid includes

- **Interstitial fluid**
 - Present between the cells
 - Approximately 80% of ECF
- **Plasma**
 - Present in blood
 - Approximately 20% of ECF
- **Also includes**
 - Lymph
 - synovial fluid
 - aqueous humor
 - Cerebro spinal fluid
 - sweat
 - seminal fluid
 - tears
 - pleural fluid

Body Fluid Compartments



(a) Distribution of body solids and fluids in an average lean, adult female and male

(b) Exchange of water among body fluid compartments

Barriers separate ICF, interstitial fluid and plasma

- Plasma membrane
 - Separates ICF from surrounding interstitial fluid
- Blood vessel wall
 - Separate interstitial fluid from plasma

Composition of body fluids

- Organic substances
 - Glucose
 - Amino acids
 - Fatty acids
 - Hormones
 - Enzymes
- Inorganic substances
 - Sodium
 - Potassium
 - Calcium
 - Magnesium
 - Chloride
 - Phosphate
 - Sulphate

Difference

ECF

- Most abundant cation - Na^+ ,
 - muscle contraction
 - Impulse transmission
 - fluid and electrolyte balance
- Most abundant anion - Cl^-
 - Regulates osmotic pressure
 - Forms HCl in gastric acid

ICF

- Most abundant cation - K^+
 - Resting membrane potential
 - Action potentials
 - Maintains intracellular volume
 - Regulation of pH
- Anion are proteins and phosphates (HPO_4^{2-})

● Na^+ / K^+ pumps play major role in keeping K^+ high inside cells and Na^+ high outside cell

Specialized Fluids of the Body

- Lymph
- Milk
- Cerebrospinal fluid (CSF)
- Amniotic fluid (AF)
- Sweat
- Tears

Specialized Fluids of the Body (continuous)

- Pleural fluid
- Saliva
- Synovial fluid
- Seminal fluid

Lymph

- Clear and colorless fluid
- 96% water and 4% solids
- Solids –
 - **Proteins**
 - 2-6% of solids
 - albumin, globulin, fibrinogen, prothrombin, clotting factors, antibodies, enzymes
 - **Lipids**
 - 5-15%
 - Chylomicrons
 - Lipoproteins
 - **Carbohydrates**
 - Glucose mainly
 - **NPN**
 - Urea and creatinine
 - **Electrolytes**
 - Sodium, calcium, potassium, chloride, bicarbonates

Functions of Lymph

- **Return protein from tissue spaces into blood**
- **Redistribution of fluid**
- **Removal of bacteria, toxins and other foreign bodies from tissues**
- **Maintain structural and functional integrity of tissue**
- **Route for intestinal fat absorption**
- **Transport lymphocytes**

Milk

- Secreted by mammary glands
- Complete natural food
- 83-87% water and 13-17% solids
- Solids
 - Carbohydrates
 - Lactose
 - Lipids
 - Triacylglycerols
 - Palmitic acid, myristic acid, stearic acid, lauric acid, butyric acid, oleic acid
 - Proteins
 - Casein (80%), Lactalbumin
 - Enzymes (proteases, xanthine oxidase, lysozyme), immunoglobulins
 - Minerals
 - Ca, Mg, P, Na, K, Cl
 - Vitamins
 - Fat and water soluble vitamins except Vit - C

Functions of Milk

- Milk sugar provides galactose, a structural unit for growing infant.
- In intestine, it gets metabolized to lactic acid which eliminates harmful bacteria.
- Source of protein, mineral and vitamins

Cerebrospinal fluid (CSF)

- Clear, colorless liquid formed within the cavities of brain and around spinal cord
- %100 ml CSF is formed everyday
- At any given time, there is 120-150ml CSF in the system
- CSF is completely replaced about three times a day.

Functions of CSF

- Hydraulic shock absorber
- Regulation of intracranial pressure
- Influences the hunger sensation and eating behaviors.

Pathology

Haemorrhage, with red blood cells in the CSF, may be secondary to:

- Hypertensive intracerebral haemorrhage into ventricles,
- Extension of a traumatic haematoma,
- Bleeding from vascular malformations

Pathology (continuous)

- **Meningitis** is an inflammation of the lepto meninges, usually caused by infection. Infecting organisms include bacteria, viruses, fungi and parasites. Microorganisms may reach the brain.
- **Malignant tumors** may shed cells into the CSF. Primary tumors e.g. gliomas may spread along the subarachnoid space. They are more common in ventricular than lumbar fluid.

Pathology (continuous)

- **Demyelinating diseases** may produce CSF abnormalities by several mechanisms. Products of demyelination may be present in the fluid (eg myelin basic protein); leucocytes from lesional tissue may shed into the fluid; and increased oligoclonal immunoglobulins produced by local synthesis at the site of lesional tissue may be washed into the fluid.

Amniotic Fluid (AF)

- Liquid produced by membranes and fetus
- Volume of fluid increases with gestational age
- Clear with some desquamated fetal cell and a little lipid.

Functions of AF

- Physical protection to the fetus
- Medium for exchange of various chemicals

Sweat

- Secretion of sweat gland
- Regulates body temperature by cooling and evaporation
- Sweat glands controlled by ANS, Adrenal cortical steroid - which affect the quantity of electrolyte present

Sweat

- Insensible perspiration amounts to 800-1200ml/day
- Volume of sweat produced/day during muscular exercise at elevated temperature may range from 10-14l, which may lead to water and electrolyte imbalance
- Water content of sweat varies from 99.2-99.7%
- pH – 4.7 to 7.5
- Total non protein nitrogen (urea) varies from 0.07-1% per hour during copius sweating

Sweat

- Electrolyte in sweat varies
 - Na⁺ 12.6-127 meq/l
 - K⁺ 5-32 meq/l
 - Cl⁻ 8.5-85 meq/l

Pathology of Sweat in Cystic Fibrosis

- **Cystic fibrosis (CF)** is caused by mutations in a gene encoding the cystic fibrosis transmembrane conductance regulator (CFTR) protein, resulting in dysfunctional epithelial chloride channels. Re-absorption of chloride ions is diminished or abolished in the duct of the sweat gland.
- There are more than 1200 known mutations of the CFTR gene.

Tears

- Produced by lachrymal glands
- Isotonic but becomes hypertonic due to evaporation as fluid passes over the cornea
- When the tear flow is copious, fluid is isotonic
- Under stimulus with a slow rate of tear flow, the fluid is about 25m osm hypertonic

Copius – Rapid tear flow induction

Tears

- pH – 7 to 7.6 due to loss of CO₂
- Protein content is 0.6 to 0.18g/dl with an albumin/globulin ratio of 1:5 or 2:1
- Small amount of mucin is also present
- Lysozyme – lyses the cells of a number of micro-organisms by breaking down the polysaccharides of their outer layer

Functions of Tears

- Lysozyme protects eye from infectious agents
- Lubricate the surface of the cornea
- Fill the irregularities of the corneal surface to improve optical properties
- Protects eyes from injury

Pleural fluid

Physiology

The pleural cavity is the space between the chest wall and the lungs. It is lined by two membranes and lubrication between these serous membranes is provided by a very thin layer of fluid, usually less than 10 ml in each cavity. Pleural fluid is an ultra-filtrate of plasma.

Pathology

A pleural effusion occurs when fluid formation exceeds removal resulting in accumulation of excess fluid in the pleural space.

The etiology of the accumulation is related to the underlying condition e.g.

- **Congestive cardiac failure**
- **Malignancy**
- **Infection**

Biochemical tests performed routinely on pleural fluids

- Total protein (normal range 25-35 g/L)
- Albumin
- LDH is about 0.6 of serum
- pH Normal is about 7.6
- Glucose

Note; serum levels of protein, LDH, pH, and glucose should be measured for comparison.

Saliva

Physiology

Whole saliva is a mixture of oral fluids including salivary gland secretions, cellular material and food debris. Saliva also contains molecules normally found in serum that reach the saliva.

Pathology

Saliva may be affected by systemic diseases or may reflect changes in serum concentrations of certain analyses. A reduction in salivary secretions is seen in;

- Malignancy
- Infection
- Endocrinopathies
- Exogenous substances such as drugs can be identified and measured in saliva.

Diagnostic use

The primary uses of saliva testing are in the areas of toxicology, endocrinology and infection.

Advantages of using salivary include ease of collection and storage.

Biochemical Tests (Endogenous analytes)

- Infectious Disease

Antibody detection includes Helicobacter pylori, Lyme disease, mumps and measles. Viral particles may be measured by PCR. Reducing the risk of infection for healthcare staff.

Biochemical Tests (Endogenous analytes) (Continuous)

■ Hormones

The majority of hormones enter saliva by passive diffusion along a concentration gradient across the acinar cells of the salivary gland. Such hormones are lipid-soluble (i.e. steroids). Salivary levels may represent the free or non-protein-bound hormone levels. Cortisol levels, for example, correlate well with serum levels (except in conditions of increased binding proteins such as pregnancy and during use of the oral contraceptive pill). Salivary testosterone also correlates well with serum levels and may be a useful test in research on male hypogonadism or in sports medicine.

Biochemical Tests (Exogenous analytes)

The presence of a drug in saliva is influenced by the physico-chemical characteristics of the drug molecule. Passive diffusion of small non-ionised molecules is the major mechanism by which a drug will appear in saliva. Since binding proteins do not cross the membrane due to their size, only the unbound fraction of the drug in serum is available for diffusion into saliva. The correlation of blood and serum levels with saliva levels differs depending on the structure of the drug. For acidic drugs, the equilibrium favors blood, while basic drugs are found in higher levels in saliva.

Biochemical Tests (Exogenous analytes) (Continuous)

Saliva can be used to detect and/or monitor cotinine, cannabinoids, cocaine, opioids, diazepam, amphetamines and ethanol. Ethanol is neither ionised nor protein-bound and, due to its low molecular weight and lipid solubility, rapidly diffuses into saliva. Thus the oral fluid to plasma concentration of ethanol averages about 1.

Seminal fluid

Physiology

Semen is a fluid formed at ejaculation. It is composed of spermatozoa in seminal plasma. Semen therefore is made up of secretions of all the accessory glands of the male genital tract as well as the testicular sperm component.

Continuous

The testicular contribution to semen volumetrically forms a relatively small portion of the ejaculate. Other secretions are produced mainly by the seminal vesicles and the prostate.

Pathology

The seminal plasma functions as a nutrient transport medium for the spermatozoa. Changes in one or more of the secretions that form the semen may have effects not only on the concentration of sperm in the ejaculate but also on sperm function. Reproductive failure may be the result of pathology of one of the accessory glands rather than an abnormality of sperm itself.

Biochemical Tests

Fructose

Fructose, which is the major source of glycolytic energy in spermatozoa, is produced by the seminal vesicles. A very low fructose level in the semen of an azoospermic man indicates absence of the seminal vesicles.

Biochemical Tests (continuous)

Alpha-glycosidase and Glycerylphosphoryl choline

Cases of obstructive azoospermia due to epididymal obstruction can be distinguished from non-obstructive azoospermia by the very low levels of epididymal derived alpha-glycosidase or glycerylphosphoryl choline.

Acid Phosphatase

Acid phosphatase is used as a marker for the presence of prostatic fluid.

Synovial fluid

Physiology

Synovial fluid is a colorless to light yellow highly viscous fluid which does not clot. It is found in joint cavities.

It is formed as an ultrafiltrate of plasma across the synovial membrane. Its function is to supply nutrients to cartilage, act as a lubricant to joint surfaces and to carry away waste products.

Pathology

Increased volume of synovial fluid may be the result of a variety of pathological processes. Such synovial fluids are often classified pathologically into four groups:

Non-inflammatory (e.g. Osteoarthritis, neuroarthropathy)

Inflammatory (e.g. Rheumatoid arthritis, gout)

Septic (e.g. Bacterial or fungal infection)

Haemorrhagic (e.g. Haemophilia, trauma).

Biochemical tests

The most commonly requested biochemistry test is glucose. A blood sample should be taken at the same time as the knee aspirate. The synovial fluid glucose concentration is normally no more than 0.6 mmol/l lower than the serum concentration.

Biochemical tests (continuous)

Significantly decreased synovial fluid glucose concentration indicate presence of inflammatory and/or septic joint disorders.